SYSDESIGN REFERENCE

Design a Rate Limiter

Requirements-to-architecture reference walkthrough.

Enforce per-identity budgets at the edge with bounded error under distributed concurrency.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / RATE-LIMITER

SYSDESIGN

Design a Rate Limiter

CORE IDEA

Enforce per-identity budgets at the edge with bounded error under distributed concurrency.

Reference architecture decision chain

01	02	03	04
Algorithm	Storage	Sharding	Atomicity

Decisions to defend

+ Algorithm
+ storage
+ sharding
+ atomicity
+ regional policy
+ and failure mode.

Failure modes to test

+ Counter races
+ uneven identities
+ latency at the limiter
+ and fail-open abuse.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

Design a Rate Limiter

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

PUBLISHING STATUS

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